Emerging Technology Case Study

Connecting LASSO regression to modern AI, robotics and computing

This appendix adds context from technology directions associated with Elon Musk's companies. It does not claim that this student project implements or powers those systems.



Figure 11. Conceptual application contexts: smart-grid monitoring, industrial sensor analysis, and renewable-energy forecasting. This is an illustrative composite, not a photograph of a specific installation.

1. Tesla Optimus: AI and robotics

Tesla describes Optimus as a general-purpose humanoid robot and identifies perception, navigation, balance, planning and physical interaction as relevant challenges [5]. LASSO could serve as an interpretable baseline for labelled sensor datasets, such as estimating a numerical equipment indicator from vibration, current and operating-time features. It would not control a robot or replace a safety controller.

2. Terafab: advanced AI computing hardware

Reuters reported on October 20, 2026 that the Terafab initiative associated with Tesla and SpaceX was targeting volume chip production in 2028, while describing the schedule as ambitious [6]. LASSO's connection is indirect: a model could analyse suitable manufacturing or chip-test data to identify informative variables. This repository does not design chips or use Terafab data.

3. SpaceX: satellite connectivity and computing

SpaceX has described plans involving next-generation Starlink satellites and space-based computing infrastructure [7]. These are company plans, not features demonstrated by this program. A future student exercise could evaluate LASSO on public labelled datasets to estimate communication demand or equipment-health indicators.

4. A realistic student-scale extension

Use a documented public sensor or energy dataset; compare LASSO with a simple baseline; choose lambda using a separate validation set or cross-validation; and report MAE, RMSE and R-squared on held-out data. Discuss data quality, changing operating conditions, privacy, cybersecurity, energy use and reliability. Do not connect an educational model directly to live equipment or robot controls.

References and Scope

[5] Tesla, “AI & Robotics,” official website. https://www.tesla.com/AI (accessed October 22, 2026).

[6] Reuters, “SpaceX and Tesla’s Terafab targets 2028 for volume chip production, Musk says,” October 20, 2026. https://www.reuters.com/business/media-telecom/spacex-teslas-terafab-targets-2028-volume-chip-production-musk-says-2026-10-20/ (accessed October 22, 2026).

[7] SpaceX, “Updates,” company updates on Starlink and space-based computing infrastructure. https://new.spacex.com/updates (accessed October 22, 2026).

Important scope statement. Elon Musk-associated companies are used as real-world examples of AI, robotics, satellite infrastructure and advanced computing. LASSO is a statistical regression technique that may be useful for particular data-analysis tasks; it is not the technology powering these systems as a whole. All links should be rechecked before final submission.